

To: Prof. Xizhao Wang, Editor-in-Chief, *International Journal of Machine Learning and Cybernetics* (Springer)

Dear Prof. Wang,

We are pleased to submit our manuscript, “FreqLite: A Lightweight Frequency-Decomposed Linear Model with Adaptive Reversible Normalization for Robust Long-Term Time-Series Forecasting,” for consideration as an original research article in *International Journal of Machine Learning and Cybernetics*.

Long-term time-series forecasting is a core task in engineering applications spanning energy systems, traffic management, weather monitoring, and industrial process control. Our work directly targets a practical gap: existing accurate forecasters (Transformer-based) are too resource-intensive for constrained deployment environments, while cheap linear models have limited representational capacity. FreqLite bridges this gap with a principled frequency-domain architecture that is both highly accurate and exceptionally resource-efficient.

The manuscript makes three contributions. First, **FreqLite**, an ultra-lightweight, channel-independent frequency-decomposed linear forecaster: a learnable partition-of-unity spectral filter splits the input signal into frequency bands modeled by dedicated linear heads. Evaluated across seven datasets (including high-variate Electricity with 321 channels and Traffic with 862 channels) against eight baselines, FreqLite at long lookback ($L=336$) attains the best overall average test MSE and the most per-setting wins of any model, **outperforming iTransformer (ICLR 2024) by 4.3% and TimesNet (ICLR 2023) by 23.2%** in paired per-cell MSE—while using $13\times$ less peak GPU memory and $18\times$ less training time than TimesNet—all on a single 4 GB consumer laptop GPU. These improvements are statistically significant under paired Wilcoxon tests ($p < 10^{-4}$). Second, **Adaptive Reversible Instance Normalization (A-RevIN)**, a regime-adaptive reversible normalization that strictly generalizes RevIN, engaging under non-stationarity (validated on ILI and a controlled distribution-shift stress test) and reducing to vanilla RevIN without harm on stationary data. Third, a **comprehensive, reproducible accuracy–efficiency study** on commodity hardware across 7 datasets and 8 baselines, with ablations isolating each component and a distribution-shift stress test that directly probes A-RevIN’s robustness claims.

This work is aligned with IJMLC’s focus on the hybrid development of machine learning methodology and its practical application: all experiments run on a single 4 GB laptop GPU, the complete codebase is publicly available and reproducible, and the paper reports limitations and non-wins honestly.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission; the authors declare no competing interests and received no specific funding.

Thank you for your consideration.

Sincerely,

Mirza Samad Ahmed Baig (Corresponding author)
Fandaqah, Al Khobar, Saudi Arabia
Mirza@fandaqah.com

Syeda Anshrah Gillani
Heidelberg University, Germany

syeda.gilani@stud.uni-heidelberg.de

Asher Ali
Fandaqah, Al Khobar, Saudi Arabia
CTO@Fandaqah.com